



McKinsey and Wonderful team up to deliver enterprise AI transformation from strategy to scale

Combining transformation expertise, advanced agentic AI capabilities, and a production-ready agent platform, our strategic collaboration helps ambitious organizations move out of “pilot purgatory” toward real, scaled impact for complex use cases.



April 7, 2026 - McKinsey and Wonderful have announced an strategic collaboration to help clients move from AI ambition to agentic AI deployment at scale.

The collaboration combines McKinsey's transformation expertise and QuantumBlack, AI by McKinsey's capabilities with Wonderful's enterprise agent platform and forward-deployed engineers, creating an

integrated offering that can take ambitious organizations from experimentation to production-grade deployment.

A [recent McKinsey survey](#) finds that while 79 percent of organizations are experimenting with gen AI, fewer than 10 percent have scaled AI agents. The gap isn't ambition; it's execution.

Our strategic collaboration aims to help clients change that.

Wonderful is one of the fastest-growing applied-AI scale-ups, with a singular focus on building and deploying agents in production. Its hyperlocal operating model and forward-deployed engineering teams enable rapid implementation directly inside client environments and pair well with McKinsey's emphasis on bringing joint teams to work "hands on keys" with clients.

"Deploying agentic AI across the enterprise is a complex endeavor that requires more than just great technology," says Bar Winkler, cofounder and CEO of Wonderful. "Enterprises often struggle because deployment is treated as an afterthought. Strategy is separated from execution, and production is separated from delivery. We've seen that AI only lands and scales inside an enterprise when technology and implementation are designed together from day one. That's where we're deeply aligned with McKinsey."

McKinsey and QuantumBlack also bring deep experience in rewiring organizations with technology and change management—the connective tissue that turns technical deployment into lasting business impact.

“The opportunity is to bring cutting-edge agentic AI into organizations in ways that create tangible value at scale—quickly—while reshaping how work gets done,” [says Lieven Van der Veken](#), a McKinsey senior partner and leader of QuantumBlack.

By pairing McKinsey’s global expertise with Wonderful’s forward-deployed engineers and hyperlocal delivery model—treating every market as a priority—clients benefit from truly integrated teams that provide both global scale and local execution.

Our work together is already demonstrating impact across industries. For example, in telecommunications, agentic workflows are enabling more personalized customer interactions, improving conversion, and streamlining operations.

“This strategic collaboration enables clients to move from use case design to live deployment with a single integrated platform,” says [Ben Ellencweig](#), a senior partner at McKinsey. “To win in a domain, our clients leverage an ecosystem of distinctive partners. This strategic collaboration is uniquely positioned to sit on top of hyperscalers to help midsize and legacy enterprises with complex tech stacks unlock value at scale.”

About Wonderful

[Wonderful](#) combines a powerful AI platform with best-in-class local deployment to deliver enterprise-grade AI agents to every market and every language. The platform enables enterprises to build, deploy, and optimize AI agents that serve customers and streamline workflows

across front and back office. Founded by Bar Winkler, chief executive officer, and Roey Lalazar, chief technology officer, with backing from Insight Partners, Index Ventures, IVP, Bessemer Venture Partners, and Vine Ventures, Wonderful enables enterprises to run human-grade agents in some of the world's most complex environments and use cases.

1936–1946: When technology innovated under pressure

As McKinsey celebrates its centenary in 2026, this series explores the defining technologies of each decade and the lessons they offer for today's technology leaders.



By [Chandrasekhar Panda](#)

Supports public-sector and financial-services institutions on technology modernization, digital public infrastructure, and AI-led transformation programs. Works at the intersection of strategy, governance, and advanced technology to drive system-level impact at a national and city scale



By [Henning Soller](#)

Leads work on technology architecture and large-scale IT, data, and AI transformations across public- and private-sector institutions. Advises governments and enterprises on building scalable, sovereign-grade digital and AI platforms to enable resilient, future-ready ecosystems



By [Klemens Hjartar](#)

Leads tech and advanced-analytics performance-based transformations, building capabilities at scale in large organizations in both B2C and B2B environments



By [Sven Blumberg](#)

Serves financial institutions across Europe, the Middle East, and Africa on operations and technology, with a focus on large-scale technology transformations, digital platforms, enterprise architecture, and IT modernization

April 29, 2026 - Over a century of technological progress, some of the most important advances have come during periods of crisis. The second decade of McKinsey's journey, 1936 to 1946, illustrates that principle. Forged in the crucible of a global war, the technological developments of this decade—including the first programmable computer—accelerated innovation in ways analogous to today's AI boom.

Beyond computing, this decade also saw advances in aviation, energy, and infrastructure that expanded people's understanding of what technology could do and how it could be applied. For today's technology leaders, the parallels are direct. The challenge shared by all advanced technologies, including agentic AI and quantum computing, is to move from breakthroughs to broad application.

Computing introduces programmability into machines

The development of the first programmable computers stands out as the biggest breakthrough of the decade. In 1936, Alan Turing described the concept of a universal machine, establishing the theoretical basis for programmable computing. Within a few years, that idea began to take practical form. Early computing systems such as Konrad Zuse's Z3 (completed in 1941) and the British Colossus (developed between 1943 and 1944) were designed for calculations and codebreaking. These early computers were large and costly, but they introduced a new concept. For the first time, machines could be programmed

to perform different tasks without their hardware being physically reconfigured. This separation of logic from machinery established the foundation for modern computing.

The lesson for modern technology leaders is clear: When a technology becomes programmable, its range of applications expands rapidly. AI is following a similar path. Early deployments are focused on specific use cases. But as capabilities mature, the opportunity shifts toward autonomous AI agents reinventing entire workflows and end-to-end processes across the enterprise. Capturing that value requires hard work to update data, platforms, and operating models alongside the core technology.

Aviation enters the jet age

Perhaps not surprising given the all-encompassing world war, aviation took great leaps forward during this period. The decade saw the introduction of jet propulsion, with the first operational jet-powered aircraft emerging in the early 1940s. In Germany, the Messerschmitt Me 262 entered service in 1944 as the first operational jet fighter, while in the United Kingdom, Frank Whittle's jet engine powered the Gloster Meteor. It began flying combat missions the same year. Earlier aircraft had already demonstrated the viability of powered flight, but jet engines extended their speed, range, and performance.

These jet engines, initially developed for military aircraft, required advances in materials, engineering, and maintenance that seemed impossible only years before. In the following decades, jet propulsion became the standard for commercial aviation. Aircraft such as the de Havilland Comet in the early 1950s and the Boeing 707 later in the decade brought jet travel to civilians and cracked open the era of global trade.

For today's CTOs, the trajectory is familiar. Jet propulsion did not transform aviation on day one; its impact came later when it was engineered for reliability and standardized for commercial use. The same dynamic is now playing out with AI. Many advanced capabilities are emerging in narrow use cases, but the real value will come when AI is deployed consistently across the enterprise. That shift will require advances in system design and integration. It will also require successful change management, helping people see AI as safe and useful—much as earlier generations had to become comfortable with the idea of flying in a metal tube.

Energy unlocks unprecedented power and complexity

The discovery of nuclear fission introduced a new source of energy with much higher power density than previous technologies. In 1938, chemists Otto Hahn and Fritz Strassmann first demonstrated nuclear fission, with physicists Lise Meitner and Otto Frisch providing the theoretical explanation shortly after. Within a few years, this scientific breakthrough moved rapidly from theory to application. The Manhattan Project was launched in 1942 and culminated in the first atomic bombs in 1945. These weapons altered the course of history and demonstrated both the scale of nuclear's potential and the magnitude of its risks.

In subsequent years, nuclear energy was adapted for civilian use. The first nuclear power plants began operating in the 1950s, and over time, nuclear power became part of the energy mix in many countries. What defines nuclear technology is the level of control it requires. Its benefits are substantial, but so are

the consequences of failure. Managing it safely depends on rigorous engineering, disciplined operating practices, and strong regulatory oversight.

The relevance for today's technology leaders is direct: As technologies become more powerful, the need for responsible use increases. When it comes to AI, companies that embed governance and accountability into their systems are better positioned to scale and sustain trust.

Infrastructure enables mobility at scale

The 1936–1946 period also saw the expansion of transportation infrastructure, particularly in railways and highways. In Germany, the Autobahn network was extended in the late 1930s to enable rapid travel across regions. In the United States and the United Kingdom, wartime demands drove tighter coordination of rail systems and the growth of long-distance trucking. These networks were no longer managed locally. They were built to support large-scale mobilization, moving troops, equipment, and materials quickly in a coordinated way. Rail schedules were aligned. Roads were better connected. Logistics operations were managed centrally to reduce bottlenecks. Over time, these systems became the backbone of economic activity. The innovation came from how a tightly integrated network functioned as a whole.

For today's technology leaders, the comparison to digital infrastructure is direct: Cloud platforms and data systems are today's most critical networks. CIOs must manage a growing ecosystem of technology partners while ensuring that their enterprise architectures integrate with cloud and AI platforms and that their data is ready for agentic workloads.

Rocketry extends the boundaries of what is possible

Advances in rocketry during this period pushed engineering progress into new territory. In Germany, Wernher von Braun and his team developed the V-2 rocket, which was the first long-range ballistic missile launched successfully, in 1942. Like other technological developments of this period, these early rocket systems were designed for the military and had little application beyond that context. But they demonstrated totally new capabilities that, not long after, opened up the cosmos for exploration. The launch of Sputnik in 1957 marked the beginning of the space age, followed by rapid advances in satellite technology.

Early rocket scientists would never have predicted their work would underpin global communications networks. This dynamic is common in emerging technologies that rely on engineering breakthroughs. The first applications reflect immediate needs, but as capabilities improve and costs decline, new use cases emerge and new value is created.

For today's CIOs, the progression of rocketry highlights a familiar challenge. Technologies that begin with narrow use cases can expand into a wide range of enterprise applications, often faster than architectures, security models, and operating practices can keep up.

Early promise fails to scale

Not all technologies of the decade achieved lasting adoption. Airships were once seen as a viable—and almost magical—solution for long-distance travel, particularly in the early 1930s. That promise unraveled

quickly after the Hindenburg disaster in 1937. Airplanes, while less elegant in some respects, ultimately proved more reliable and economically viable at scale. Airships could perform under controlled conditions, but they frayed under the more brittle demands of commercial deployment.

This paradox remains relevant today. AI has shown strong results in pilots, but few companies have successfully scaled it to deliver lasting value. Our research illustrates that gap: Some 88 percent of companies report regular AI use in at least one business function, but only one-third have started scaling their AI programs.

For technology leaders, the implication is practical. Scaling requires more than technical performance. It also requires strengthening the underlying operating muscle. Rushing to deploy agentic AI without adopting a flexible product and platform model could risk a Hindenburg-level resiliency failure. Success with cutting-edge technology depends on understanding how systems perform under real-world variability and whether the broader operating model can support sustained use.

A leadership lesson endures

During the 1936–1946 period, many of the most important technologies were developed through coordinated efforts rather than isolated invention. With the world at war, scientists and engineers mobilized to push progress to its outer limits in pursuit of strategic advantage. Progress depended on the ability to integrate disciplines and ensure that systems functioned under very difficult real-world conditions. Scaling AI requires the same discipline. Organizations that get this right will be the ones that turn today's breakneck technological progress into controlled and sustained advantage.

Chandrasekhar Panda is a partner in McKinsey's Riyadh office, **Henning Soller** is a partner in the Frankfurt office, **Klemens Hjartar** is a senior partner in the Copenhagen office, and **Sven Blumberg** is a senior partner in the Istanbul office.

McKinsey Technology



Shaping a new era: CIOs on tech and business collaboration

At McKinsey Technology's annual CIO Conference, CIOs stressed that the next wave of technological advancement and automation will require close collaboration across the business.



By [Anna Wiesinger](#)

Advises public- and private-sector clients, with a focus on digitalization processes and transformation management for education, employee, job market, and manager development



By [Henning Soller](#)

Leads work on technology architecture and large-scale IT, data, and AI transformations across public- and private-sector institutions. Advises governments and enterprises on building scalable, sovereign-grade digital and AI platforms to enable resilient, future-ready ecosystems



By [Sven Blumberg](#)

Serves financial institutions across Europe, the Middle East, and Africa on operations and technology, with a focus on large-scale technology transformations, digital platforms, enterprise architecture, and IT modernization



By [Martina Gschwendtner](#)

Serves tech companies and their investors on incorporating gen AI, robotics, and quantum technologies



By [Thao Dürschlag](#)

Serves clients in the banking sector on AI transformations, focusing on governance and operating model

April 22, 2026 - A year ago, most gen AI use cases existed as proofs of concept or early-stage ideas not yet fully implemented in any organization. But the technology is maturing rapidly, and organizations are now deploying agents and gen AI solutions at significant scale: 28 percent of top-performing companies plan to increase their technology budgets by more than 10 percent in 2026 specifically to scale agentic AI systems that autonomously plan and act across workflows.

At McKinsey Technology's annual CIO Conference, we convened CIOs from all major industries to discuss how this AI revolution is shaping a new era technologically and from a business perspective. Four main themes emerged.

Speed has become a more significant differentiator than size

In the past, organizations have often harnessed economies of scale to become more competitive, the understanding being that the larger the organization, the easier it is to increase efficiency. But with the massive proliferation of agents and the possibility to automate vast amounts of work, scale can be achieved far more readily with automation than through organizational size alone. Large organizations may struggle to move quickly and innovate, but the age of agents requires quick decision-making and clear guardrails to build faster innovation cycles. Smaller organizations may thrive under these conditions. [McKinsey research](#) shows that top-performing companies are adopting [product and platform operating models](#) where decisions happen within days instead of months.

It is no longer sufficient to pursue a technology agenda in isolation

Traditionally, technology functions handled digitization. Business and technology teams worked through dedicated interfaces, structured around product owners and business owners, to translate operational needs into technical requirements, while the rest of the organization continued to run as before. But today, automation is extending beyond the technology function to include every team on the business side, with agents working alongside employees to automate tasks and streamline workflows. As a result,

46 percent of top performers now [cocreate strategy across business and tech teams](#) iteratively throughout the year—a figure that has doubled since 2023.

[McKinsey Global Institute research](#) indicates the future of work will be a partnership among people, agents, and robots, all powered by AI. Today's technologies could theoretically automate more than half of current work hours, making the question of how organizations prepare their people as consequential as the technology decisions themselves.

This paradigm shift requires a fundamental restructuring of both business and technology teams. It also requires reskilling at scale. About half of top performers are already investing in reskilling their own workforces to deliver the next wave of change—setting up operating models that integrate human and agentic work and ensuring teams across the enterprise are equipped to work with agents. An environment where human beings and agents can collaborate and thrive requires incorporating the strengths of human beings while focusing on the opportunity ahead and mitigating risks.

Autonomous capabilities are now deeply embedded in business processes

AI solutions and robots have historically automated parts of existing processes—such as chatbots resolving routine customer queries before passing complex cases to a human agent—with manual workarounds as an option in case technology failed or produced incorrect outputs. But agents are now able to manage numerous steps and interact with each other.

In customer service, for example, agents can classify tickets, identify root causes, propose resolution paths, and autonomously resolve frequent low-complexity issues end to end. These operating model shifts are already happening, with 23 percent of organizations reporting scaling AI agents in at least one business function and an additional 39 percent actively experimenting.

These new operating models no longer allow for manual fallbacks. Therefore, organizations must harden their technology solutions—reducing vulnerabilities and building resilience (for example, by removing unnecessary software), regularly installing security updates, and introducing redundancy so that operations can continue if individual components, such as cloud providers, fail.

Partnering within the ecosystem

Scaling agentic AI extends beyond the boundaries of any single organization. As agents increasingly operate across shared platforms, cloud infrastructure, and third-party systems, the interfaces between organizations become as consequential as the internal workflows they connect. Technology leaders can no longer treat vendor and partner relationships as transactional. Instead, codeveloping solutions end to end with cloud providers, technology vendors, and implementation partners—as well as aligning on architecture, governance, and risk from the outset—will be essential to capturing the full value of the agentic era. Organizations that get this right will find that these partnerships become a source of competitive advantage, not just a procurement necessity.

In this new era of automation, people across functions could have the opportunity to do more meaningful and higher-value work. Realizing this prospect and capturing the full benefits of AI transformations requires establishing smaller, more agile teams with a sustained focus on upskilling employees to work effectively with large-scale agentic systems.

Across all discussions at the conference, one idea was constant: The next wave of technological advancement cannot be achieved through technological excellence alone. Connecting people across disciplines will be critical, and close collaboration among business and technology leaders, domain experts and data scientists, and hardware and software capabilities will ultimately determine whether organizations can convert the potential of agentic AI into trillions of dollars in sustainable annual impact. We are already looking forward to continuing these conversations at next year's conference.

[Anna Wiesinger](#) is a partner in McKinsey's Düsseldorf office, [Henning Soller](#) is a partner in the Frankfurt office, [Sven Blumberg](#) is a senior partner in the Istanbul office, and **Martina Gschwendtner** is a consultant in the Munich office, where **Thao Dürschlag** is an associate partner.

The authors wish to thank Emily Hoppe, Hannah Hoehnke, Hannah Mayer, Kimberly Beals, Marie Heller, and Sophie Westphal for their contributions to this blog post.

McKinsey Technology



1926–1936: When technology started to scale

As McKinsey celebrates its centenary in 2026, this series explores the defining technologies of each decade and the lessons they offer for today's technology leaders.



By [Chandrasekhar Panda](#)

Supports public-sector and financial-services institutions on technology modernization, digital public infrastructure, and AI-led transformation programs. Works at the intersection of strategy, governance, and advanced technology to drive system-level impact at a national and city scale



By [Henning Soller](#)

Leads work on technology architecture and large-scale IT, data, and AI transformations across public- and private-sector institutions. Advises governments and enterprises on building scalable, sovereign-grade digital and AI platforms to enable resilient, future-ready ecosystems



By [Klemens Hjartar](#)

Leads tech and advanced-analytics performance-based transformations, building capabilities at scale in large organizations in both B2C and B2B environments



By [Sven Blumberg](#)

Serves financial institutions across Europe, the Middle East, and Africa on operations and technology, with a focus on large-scale technology transformations, digital platforms, enterprise architecture, and IT modernization

April 8, 2026 - As we reflect on 100 years of technological progress, we see more than a catalog of inventions; we see ideas that evolved into systems. The first decade of McKinsey's journey, from 1926 to

1936, represents a critical inflection point. This was a period in which modern technology began to transition from isolated innovation to industrialized, networked deployment. This was the decade when technology stopped being episodic and started becoming systemic.

Across aviation, manufacturing, media, and fundamental science, the most consequential advances of the period were not singular eureka moments. Quietly at first, and then faster at the end of the decade, these new technologies altered how organizations, economies, and societies functioned. Individual technological advances transitioned into scalable systems. For technology leaders navigating today's transitions—such as shifting AI pilots to full-scale agentic platforms or migrating cloud applications to cloud-native operations—the parallels are striking.

Heroic feats become dependable systems

Aviation provides the clearest illustration of how technology evolved between 1926 and 1936 and is the closest parallel to today. By the mid-1920s, powered flight was no longer a novelty. Aircraft had already crossed oceans and captured public imagination. What changed most about aviation during these ten years was not just that they could travel faster and farther, but that they could so reliably. This shift happened when commercial aviation emerged as an industry. Airline companies learned to operate aircraft safely and predictably. This required advances far beyond airframes and engines, including standardized maintenance practices, pilot training programs, air traffic procedures, route planning, weather forecasting, and regulatory oversight. Airlines began to resemble logistics companies rather than daredevil clubs.

The lesson for modern technology leaders is enduring: The moment a technology becomes strategically important is the point at which its surrounding operating model matters more than its core performance metrics. Many AI deployments today sit precisely at this inflection point. The models exist. The demos impress. What separates leaders from laggards is not how many AI experiments they have tried or how far they have pushed the models but the ability to industrialize deployment. High-performing companies have surpassed AI pilots to transform into agentic organizations. They consider governance, monitoring, talent and operating models, and accountability for outcomes as important as technology deployment. They use AI to reconfigure entire ways of working and to equip their teams with superpowers.

Aviation's first commercial decade reminds us that scale is rarely achieved by brilliance alone. It is achieved by the hard work of systems thinking.

Manufacturing learns to think in processes, not products

During the 1926–36 decade, a similar shift occurred in manufacturing. The automobile was no longer new in the late 1920s. What was new was the maturation of operational discipline: standardized parts, assembly-line optimization, supply chain coordination, and statistical approaches to quality.

This period cemented the idea that competitive advantage could be engineered into processes, not just products. Cost curves began to bend downward not because cars became radically different but because companies learned how to produce them consistently and efficiently at scale.

This same logic underpins every modern technology transformation. Cloud computing, for example, did not win because virtual machines were conceptually revolutionary. It won because hyperscalers

embedded operational excellence into provisioning, reliability, security, and pricing. They turned compute power into an industrial utility.

CTOs looking to generate measurable value from their technology migrations today would do well to remember that technology for technology's sake does not deliver ROI. Instead, true value comes from tying technology to clear organizational processes mapped to business goals. Managing deployment pipelines, cybersecurity, vendor orchestration, change management, and financial accountability are what make technology transformations stick.

Media becomes a platform business

The 1926–36 decade also paved the way for media outlets to become true technology platforms. Radio networks expanded rapidly, creating shared national experiences in real time. Film transitioned from silent pictures to synchronized sound, triggering massive reinvention across production, distribution, and talent ecosystems.

What made these technologies powerful was not merely their reach, but their economics. Advertising models, sponsorships, and distribution agreements emerged that allowed content to be monetized at scale. Technical and contractual standards enabled interoperability across regions and devices. In retrospect, radio networks look like early prototypes of today's digital platforms: centralized infrastructure supporting decentralized content creation, governed by evolving rules and incentives.

For today's CTOs, the relevance of yesterday's media landscape to today's technology road map is clear. Platform dynamics—such as who controls interfaces, who captures data, and who sets standards—often

matter more than the underlying technology stack. The winners are rarely those who invent first; they are those who architect ecosystems.

The quiet power of foundational science

Not all of this decade's most important advances were visible at the time. In physics, the development of quantum mechanics altered humanity's understanding of matter and energy. Its immediate applications were limited, and its implications were opaque to most business leaders of the era. A hundred years later, quantum computing is moving from a bold idea to a multibillion-dollar industry with real measurable value. In fact, every technology that defines modern life, including semiconductors, lasers, and sensors, rests on scientific principles established in the decade after 1926.

This is a reminder that companies that optimize for near-term ROI risk starving themselves of future innovation. For companies looking to transform into agentic organizations, this means taking a portfolio approach that balances technology investment. First-movers fund applied AI engineering with selective investment in foundational capabilities such as research and skill building. These investments may not monetize quickly, but they can reshape competitive advantage over time.

When elegance fails to scale

The decade of 1926 to 1936 also offers cautionary tales. Airships, or dirigibles, represented a technologically elegant solution to long-distance travel. Yet safety concerns and economic constraints ultimately doomed commercial blimps. This pattern repeats throughout technological history. Elegance without resilience rarely survives contact with scale.

High-performance companies know that pilots that perform beautifully in controlled environments can collapse under real-world complexity—and nowhere is this truer than with AI. Experimentation with agentic platforms leads to learning and innovation, but it's also a messy process to engage with a brand new technology evolving at light speed. To make this type of experimentation scalable and impactful, companies need [responsible AI guardrails](#), robust security, and clear governance. These are prerequisites for durability.

A leadership lesson that endures

The 1926–36 decade was when technology stopped being something organizations *used* and became something organizations had to *manage*. Strategy, operations, governance, and talent all became inseparable from technical capability. The leaders who succeeded in this era were not necessarily the greatest inventors. They were the ones who understood how to embed technology into institutions. They built operating discipline and made innovation repeatable by creating governance frameworks.

That same reality defines the modern CIO role. Today's technology tools are different from 100 years ago. The scale of disruption is larger, and the potential benefits—and risks—are greater. But the work of turning raw promise into measurable value remains unchanged.

Chandrasekhar Panda is a partner in McKinsey's Riyadh office, **Henning Soller** is a partner in the Frankfurt office, **Klemens Hjartar** is a senior partner in the Copenhagen office, and **Sven Blumberg** is a senior partner in the Istanbul office.

McKinsey Technology



Ushering in a new era of trusted AI

Artificial intelligence will compel organizations to elevate their compliance. Six levers hold the key.



By [Henning Soller](#)

Leads work on technology architecture and large-scale IT, data, and AI transformations across public- and private-sector institutions. Advises governments and enterprises on building scalable, sovereign-grade digital and AI platforms to enable resilient, future-ready ecosystems



By [Anselm Ohme](#)

Serves clients across industries on AI, tech regulation, and cybersecurity transformations



By [Fares Darwazeh](#)

An alumnus of McKinsey's Riyadh office



By [Thao Dürschlag](#)

Serves clients in the banking sector on AI transformations, focusing on governance and operating model

March 30, 2026 - Each week seems to bring another announcement about new AI tools with the potential to reshape how organizations function. This progress has been accompanied by the growing recognition of the importance of trusted AI, defined by the OECD as corresponding solutions that are reliable, transparent, fair, resilient, and accountable.¹ The concept of trusted AI brings together technical robustness and security with ethical and social responsibility.

The rapid advancement and widespread adoption of AI have made data privacy and AI regulation critical topics on the global regulatory agenda. Yet the implementation of these guidelines and the regulatory landscape itself have become increasingly complex over the past several years.

Despite the higher regulatory burden, many organizations continue to depend on manual processes to implement regulatory and AI-related controls, adding layers and cost. As McKinsey's 2025 analysis of regulatory technology (RegTech) shows, financial institutions relying on manual compliance systems often fulfill only a fraction of their obligations, leaving them at higher risk of penalties and operational inefficiencies.

There is a high degree of difficulty in striking the right balance in compliance. For example, a US-based bank's legacy system met just 75 percent of requirements before the adoption of an automated RegTech solution streamlined data mapping, raising compliance to above 95 percent.

Given these competing challenges, companies should consider viewing compliance not as a cost but as an enabler of AI scale. This approach entails adopting a more streamlined, scalable, and digital approach to regulatory adherence. Once organizations overcome some common obstacles in the evolving regulatory environment, six levers drawn from best practices can help organizations elevate their AI compliance efforts. Companies that get it right can not only unlock faster AI deployment but also reduce their risk.

Understanding the challenges

The widespread adoption of AI chatbots and voice bots, augmenting human behavior with human voice, the rise of personalized, tailored marketing, and rapid advancements in generative AI in new products and services offered to customers have prompted regulators to establish clear guardrails, with the aim of providing customers and employees with greater control over their data and preventing fraud.

Regulations such as the General Data Protection Regulation (GDPR) and the European Union AI Act (as part of the EU Data Governance Act), as well as cybersecurity frameworks such as the Digital Operational Resilience Act (DORA), have emerged as key pillars. Similar initiatives have been introduced globally, including in Brazil, Thailand, and Türkiye, as well as by individual US states, such as the California Consumer Privacy Act (CCPA).

Despite the efforts of policymakers to provide clear guidance on compliance, the relationship between various regulations remains confusing. The result is that no universal standard exists to guide companies in simultaneously managing data privacy, verifying trusted AI, and promoting resilience. Conflicting

requirements across jurisdictions further complicate compliance; even within regions such as Europe, interpretations of regulations such as the GDPR vary significantly between countries.

Organizations have tended to respond to these regulatory demands in an ad hoc manner. Often, impending compliance deadlines spur companies to reactively implement transparency measures and controls. For example, some organizations have deployed surveys and interviews to compile process repositories. This fragmented and inconsistent documentation undermines the organization's ability to maintain accurate, updated compliance records and controls as well as alignment across business units. Similarly, controls—such as those governing the use of personal data or adherence to the principles of the EU AI Act²—are frequently implemented as manual checks. Last, companies can fail to assign responsibility for compliance, which can end up distributed across multiple departments, such as legal, IT, and compliance.

Adopting sustainable solutions for digital and AI compliance

To consistently achieve digital and AI compliance with emerging regulations, organizations should consider six levers.

1. Centralize compliance within existing teams

Compliance efforts should be incorporated into established processes and overseen by teams with a holistic view of the organization. For instance, data privacy teams could also take responsibility for compliance with the EU AI Act, or teams in charge of complying with ISO 9001 (a standard for quality management systems) could oversee the creation of process repositories. Centralizing oversight can eliminate duplicate and inconsistent documentation—for example, separate process representations for ISO 9001 and GDPR—thereby improving efficiency and reducing complexity. Such approaches are increasingly visible across industries, especially in banking, where we have seen several major institutions introduce centers of excellence that orchestrate data and AI responsibility among IT, business, and risk functions.

While execution necessarily spans legal, IT, data, and business functions, organizations should consider anchoring AI compliance accountability within the risk or compliance function, given its mandate for enterprise-wide control, regulatory interpretation, and independent oversight.

2. Digitalize controls through the use of legal tech tools

Many organizations are in the early stages of adopting legal tech tools, with factors such as cultural resilience, data security concerns, and skills gaps impeding integration. Still, these tools hold significant potential. As one example, generative AI tools can automate policy adherence checks, flagging exceptions for further review. Tools that identify personally identifiable information (PII) can reduce the time and errors associated with manual processes. A comprehensive review of existing controls, with a

focus on automating repetitive tasks such as data retention management, can not only streamline compliance efforts but also enhance accuracy and consistency.

3. Simplify overarching AI governance structures

Many organizations have overly complex governance setups, which hinder compliance efforts. This is especially true for multinational corporations with numerous subsidiaries, joint ventures, and regional entities. Simplifying policy frameworks across legal entities and consolidating oversight under a single chief compliance officer can dramatically reduce complexity and facilitate data transfer and compliance checks. Organizations should consider moving toward a federated model in which the risk or compliance function provides enterprise-level governance and challenge, while legal, IT, data, and business teams own implementation across the AI life cycle.

Regulators themselves are increasingly acknowledging the need for simplification and encouraging companies to streamline their approaches to compliance. For instance, the European Commission's Digital Omnibus initiative explicitly aims to reduce the administrative burden and overlapping requirements across digital, AI, and data laws, creating a more coherent and predictable regulatory landscape. Similarly, international policy bodies such as the G7 and OECD have emphasized the importance of harmonizing AI and data governance frameworks globally to minimize fragmentation and ease cross-border compliance.

4. Design change management processes for all employees

To facilitate the integration of compliance into daily operations, organizations would need to prioritize change management for their workforce. Targeted training and awareness programs for employees can reinforce the importance of data privacy, ethical AI, and regulatory compliance in their daily roles. Tailored, role-specific guidance for functions such as IT, legal, and operations supports alignment with compliance requirements across the organization. For instance, we have seen a bank achieve a successful technological transformation by dedicating 34 percent of its investment to structured change management efforts.

In addition, fostering a culture of accountability and ethical decision-making, supported by a clear commitment and communication from leadership, helps weave these principles into the organizational fabric. Empowering employees to understand and embrace these responsibilities creates a workforce that actively supports compliance and resilience efforts.

5. Build resilient and ethical AI systems across the value chain

To ensure both resilience and ethical behavior, organizations must go beyond compliance controls to focus on the technical foundations of their AI systems. This lever includes designing technology

architectures that are robust, scalable, and capable of withstanding disruptions such as cyberattacks, system failures, or regulatory changes.

Equally important is the development of ethical AI systems, which requires organizations to implement frameworks that promote transparency and fairness while mitigating bias. Such frameworks include the OECD AI Principles and the IEEE P7001 transparency standard. Critical components of this effort are regular audits of AI models, explainability mechanisms, and adherence to ethical guidelines throughout the AI life cycle. By addressing these technical aspects, organizations can build AI systems that both meet regulatory requirements and earn the trust of customers, employees, and other stakeholders.

6. Undertake continuous monitoring and improvement

Compliance is not a static goal but an ongoing process that requires continuous monitoring and improvement. Organizations can use AI tools that can proactively identify and address potential risks in real time. Establishing feedback loops with employees, customers, and other stakeholders provides valuable insights into the effectiveness of compliance practices and opportunities for refinement. Periodic reviews of policies, controls, and governance structures help them keep pace with evolving regulations and business needs to maintain operational resilience over time.

Next steps for implementation

To fully harness these levers, organizations could take the following steps:

- *Assess the current state.* Conduct a thorough benchmarking exercise (such as [McKinsey's AI Trust Maturity Model](#)) to compare the organization's current setup against industry best practices. A full evaluation of the organizational structure, policy frameworks, and control mechanisms can highlight gaps between an organization's AI impact aspirations and its AI trust and governance readiness.
- *Identify automation opportunities.* Collect and review all relevant controls, along with their current implementation methods, to flag opportunities for automation. Organizations should prioritize controls that are time-intensive, error-prone, or repetitive. Candidates for automation are processes that occur on a defined schedule (such as monthly reviews, quarterly audits, or continuous monitoring) rather than one-off checks.
- *Develop a unified program.* Translate insights from the benchmarking and automation assessments into a comprehensive program, with a central oversight team in the lead. This program should focus on simplifying governance structures, digitalizing controls, and standardizing compliance efforts across the organization.

The era of trusted AI demands a fundamental shift in how organizations approach compliance. The six levers offer a path for organizations to transition from reactive, manual compliance processes to a proactive, digital-first approach that reflects the demands of the new regulatory landscape. These

changes can also reduce the burden of compliance and position organizations to thrive in an increasingly regulated digital environment.

Henning Soller is a partner in McKinsey's Frankfurt office, **Anselm Ohme** is a consultant in the Berlin office, **Fares Darwazeh** is an alumnus of the Riyadh office, and **Thao Dürschlag** is an associate partner in the Munich office.

¹ "AI principles," OECD, accessed February 20, 2026.

² The act sets requirements for all companies that use, deploy, or integrate AI systems. The obligations center on governance, accountability, and oversight: ensuring the system's performance, addressing bias and unfairness, maintaining human oversight, and being able to demonstrate compliance.

State of AI trust in 2026: Shifting to the agentic era

Findings from McKinsey's 2026 AI Trust Maturity Survey reveal progress in trust maturity, alongside persistent gaps in strategy, governance, and risk management.



By [Gabriel Morgan Asaftei](#)

Develops applications powered by artificial intelligence and machine learning for real estate, retail, and financial services companies



By [Roger Roberts](#)

Advises clients in a broad range of industries as they address their toughest technology challenges—and make the most of emerging opportunities



By [Abby Sticha](#)

Advises clients in the banking and technology, media, and telecommunications industries on AI and agentic AI strategy, solution development and building, and AI trust



By [Cécile Prinsen](#)

Coleads our McKinsey AI Trust service line in Europe, the Middle East, and Africa, and leads our work in data and technology risk with financial institutions

March 25, 2026 - AI adoption is accelerating rapidly, with organizations moving beyond experimentation toward scaled deployment of gen AI and, increasingly, agentic AI across core business functions. But as AI systems take on greater autonomy—making recommendations, triggering actions, and interacting with other systems—the consequences of failure grow materially. In this environment, AI trust and the responsible AI (RAI) practices that enable trust are no longer a tangential concern but a foundational requirement for realizing the full potential of the technology.

Trust underpins two critical outcomes. First, it enables organizations to realize value from AI investments by supporting sustained adoption and integration into core workflows. Second, it is essential if organizations are to successfully manage an expanding and evolving risk landscape. In the age of agentic AI, organizations can no longer concern themselves only with AI systems *saying* the wrong thing; they must also contend with systems *doing* the wrong thing, such as taking unintended actions, misusing tools, or operating beyond appropriate guardrails.

To understand how organizations are responding to this shift, McKinsey conducted the 2026 AI Trust Maturity Survey. Taken between December 2025 and January 2026, the survey gathered responses

from approximately 500 organizations across industries and regions, with respondents who hold direct responsibility or expertise in AI governance, risk management, or AI investment decisions. Their responses were assessed using the McKinsey AI Trust Maturity Model, a framework based on five dimensions of RAI: strategy, risk management, data and technology, governance, and, new this year, agentic AI governance and controls, reflecting the growing importance of governing increasingly autonomous AI systems (Exhibit 1). We assess RAI maturity across four levels, from the development of foundational RAI practices to the implementation of a comprehensive and proactive program.

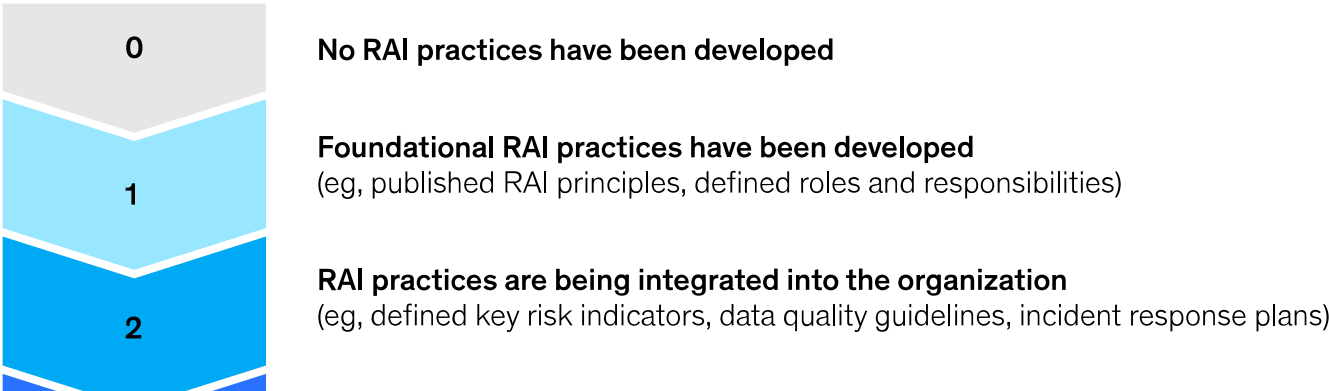
Exhibit 1

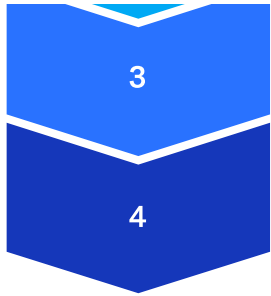
The AI Trust Maturity Model is a responsible AI framework that encompasses five dimensions.

Dimensions of responsible AI (RAI)

Strategy	Risk management	Data and technology	Governance	AI agent governance and controls
Defined vision and principles for AI or agentic AI governance aligned with client's values, strategic goals, and regulatory standards	Proactive monitoring and steps to reduce risks (incl model risk management, third-party solution auditing) through review and challenge (eg, red teaming, war gaming)	At-scale checks to instill confidence in data quality, which underlie AI solutions and ensure readiness of supporting AI governance tooling and development protocols	Talent, governance, and team structure in place to maintain AI governance across all client activities (incl processes to identify, assess, and mitigate AI risks); plan for AI-led shifts in skills, training programs, change management, and communication	Structures, oversight models, and technical guardrails to ensure that AI agents operate within defined boundaries (including agent registration and documentation, testing and evaluation of agent behavior and tool use, defined human-in-the-loop oversight, and real-time monitoring and controls)

Levels of RAI maturity





Steps are being taken to develop all necessary RAI practices

(eg, documented RAI processes, compliance assessments)

Comprehensive and proactive RAI practices are in place

(eg, allocated resources and budget, AI governance forum, real-time monitoring)

McKinsey & Company

At a glance: Key insights from the 2026 AI Trust Maturity Survey

The survey uncovers ten key insights about AI trust, which fall across three themes: the current state of AI trust, emerging risks and challenges, and how organizations are responding to close gaps and enable scale.

State of AI trust today

Despite improved maturity on the whole, significant variance still exists, but companies are increasingly recognizing the importance of RAI investment:

1. RAI maturity continues to improve, yet strategy, governance, and agentic AI controls lag behind, with only about 30 percent of organizations reaching a maturity level of three or higher in these dimensions.
2. RAI maturity varies by industry and region: Asia–Pacific leads globally, and technology, media, and telecommunications and financial services outperform other sectors.
3. Investment in RAI is strongly associated with higher RAI maturity and realized value.

Emerging risks and challenges

Increased adoption or consideration of adoption is leading to more risks and challenges as well as declined confidence in handling these risks:

4. Security and risk concerns are the top barrier to scaling agentic AI.
5. Inaccuracy and cybersecurity remain the most frequently cited AI risks as adoption expands.
6. Active mitigation lags behind risk awareness across nearly every AI risk category.
7. AI incident frequency remains stable, but confidence in organizational response has declined.

How enterprises and organizations are responding

As organizations implement RAI, they're facing barriers and differences in maturity while RAI is increasingly becomes a business enabler:

8. Knowledge and training gaps are the leading barrier to RAI implementation.
9. Organizations with explicit accountability for RAI achieve higher maturity scores than those without clear accountability.
10. AI trust is increasingly viewed as a business enabler rather than a compliance exercise.

The state of AI trust today: Scrambling to keep up with the pace of change

The AI Trust Maturity Survey provides a detailed view into how organizations are approaching issues of trust and responsibility. Below, we examine the above ten insights in detail, looking at overall maturity levels, how progress differs across industries and regions, and how investment in RAI efforts relates to both trust capabilities and realized AI value.

Insight 1: RAI maturity continues to improve, yet strategy, governance, and agentic AI controls lag behind

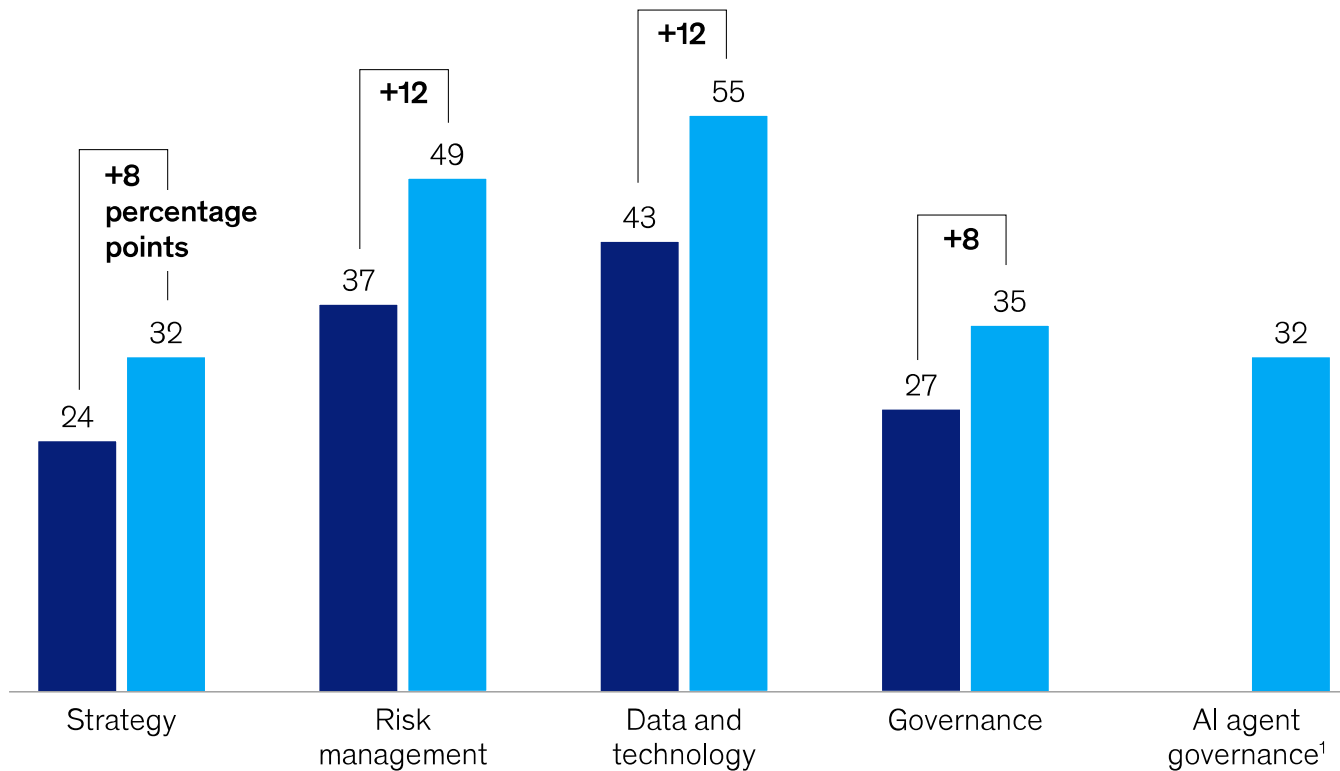
The average RAI maturity score increased to 2.3 in 2026, up from 2.0 in 2025. However, only about one-third of organizations report maturity levels of three or higher in strategy, governance, and agentic AI governance (Exhibit 2). This imbalance suggests that while technical and risk management capabilities are advancing, organizational alignment and oversight structures are struggling to keep pace with the rapid expansion of AI use.

Exhibit 2

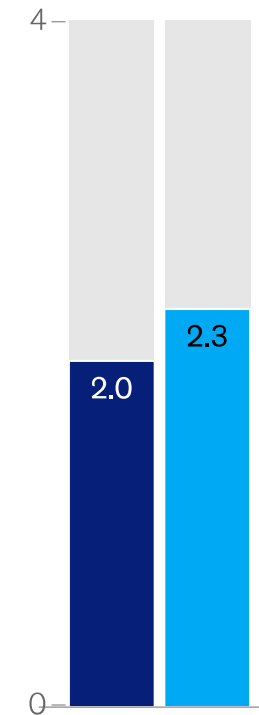
Responsible AI maturity is improving, but strategy, governance, and agentic AI governance and controls are lagging behind.

■ 2025 ■ 2026

Year-over-year change for organizations with at least level 3 maturity in responsible AI (RAI) practices, % of respondents



Overall RAI maturity score



¹Section added in the 2026 AI Trust Maturity Survey (not included in 2025 survey).

Source: McKinsey AI Trust Maturity Survey, Dec 2024–Jan 2025 (n = 759); McKinsey AI Trust Maturity Survey, Dec 2025–Jan 2026 (n = 496)

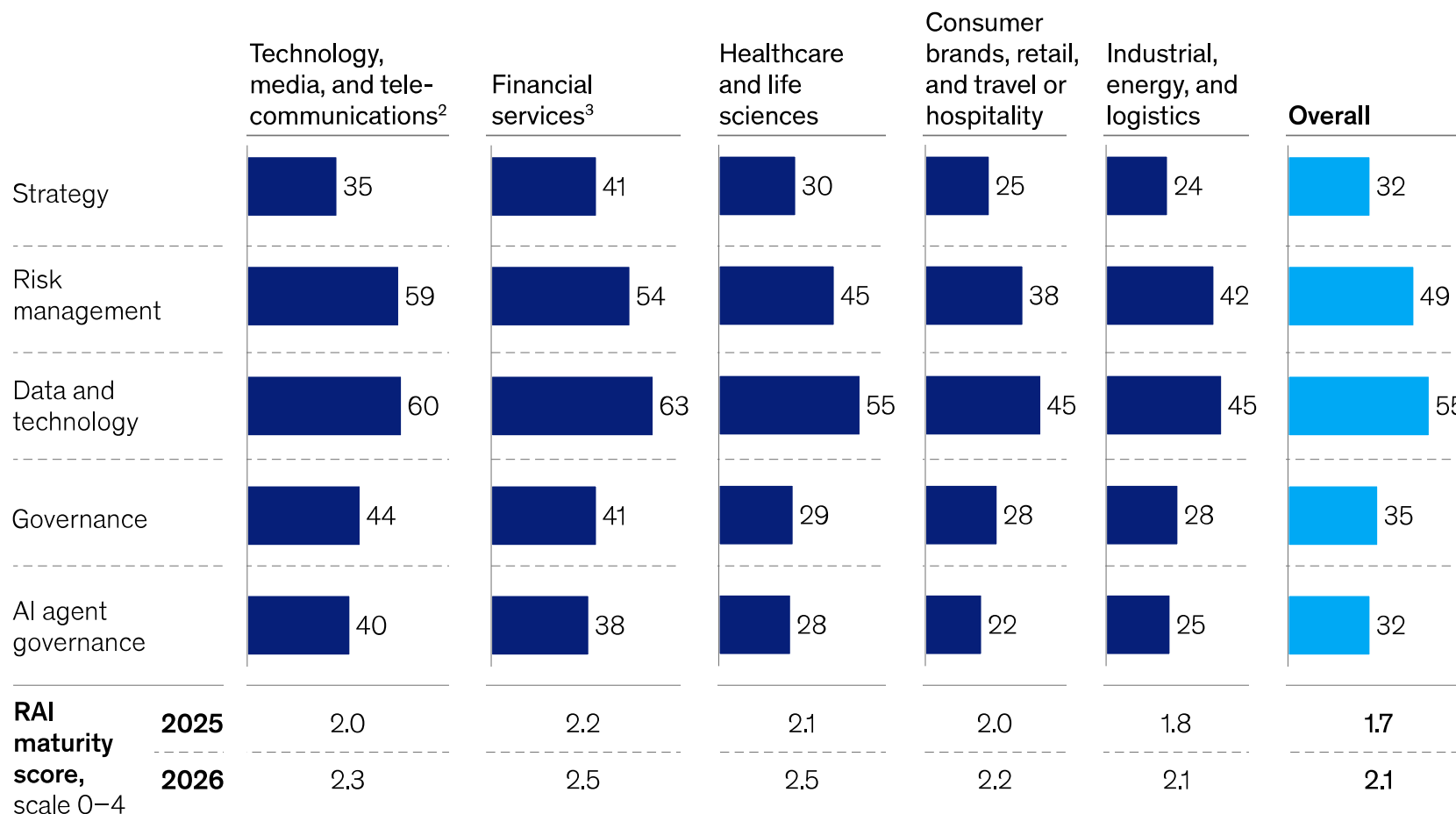
Insight 2: RAI maturity varies by industry and region

Technology, media, and telecommunications and financial services continue to lead in RAI maturity, driven by stronger risk management and data foundations (Exhibit 3). Regionally, Asia–Pacific leads overall maturity, while governance and agentic AI controls lag behind data and technology across all regions, indicating a globally consistent governance gap (Exhibit 4).

Exhibit 3

Responsible AI maturity varies by industry, with technology, media, and telecommunications and financial services leading.

Share of respondents¹ with at least level 3 maturity in responsible AI (RAI) practices, %

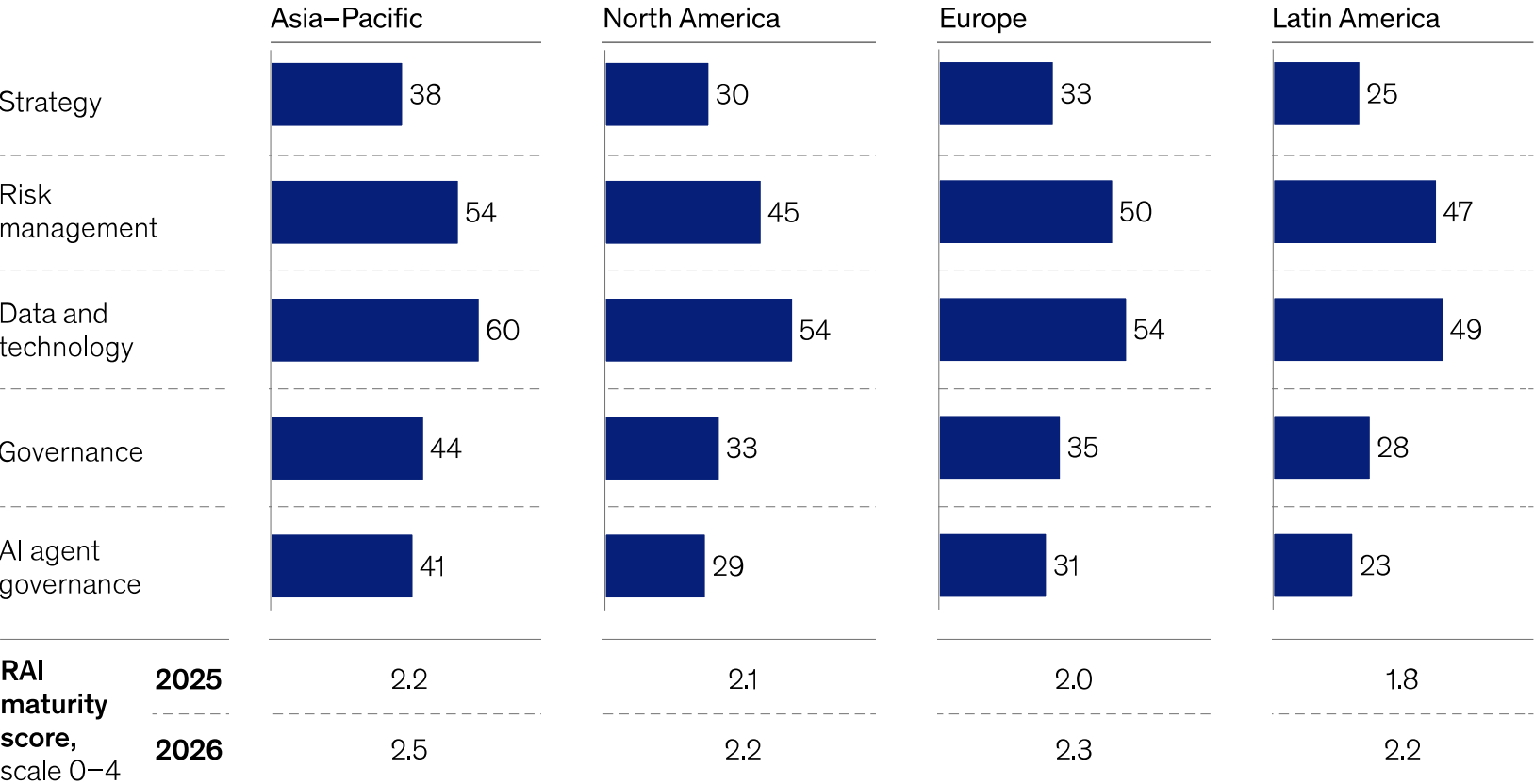


¹Technology, media, and telecommunications: n = 101; financial services: n = 146; healthcare and life sciences: n = 50; consumer brands, retail, and travel or hospitality: n = 100; industrial, energy, and logistics: n = 99. ²Includes semiconductors. ³Includes banking and payments and insurance and asset management. Source: McKinsey AI Trust Maturity Survey, Dec 2024–Jan 2025 (n = 759); McKinsey AI Trust Maturity Survey, Dec 2025–Jan 2026 (n = 496)

Exhibit 4

Responsible AI maturity varies by region, with Asia–Pacific leading overall.

Share of respondents¹ with at least level 3 maturity in responsible AI (RAI) practices, %



¹Asia–Pacific: n = 95; Europe: n = 157; North America: n = 190; Latin America: n = 43.
Source: McKinsey AI Trust Maturity Survey, Dec 2024–Jan 2025 (n = 759); McKinsey AI Trust Maturity Survey, Dec 2025–Jan 2026 (n = 496)

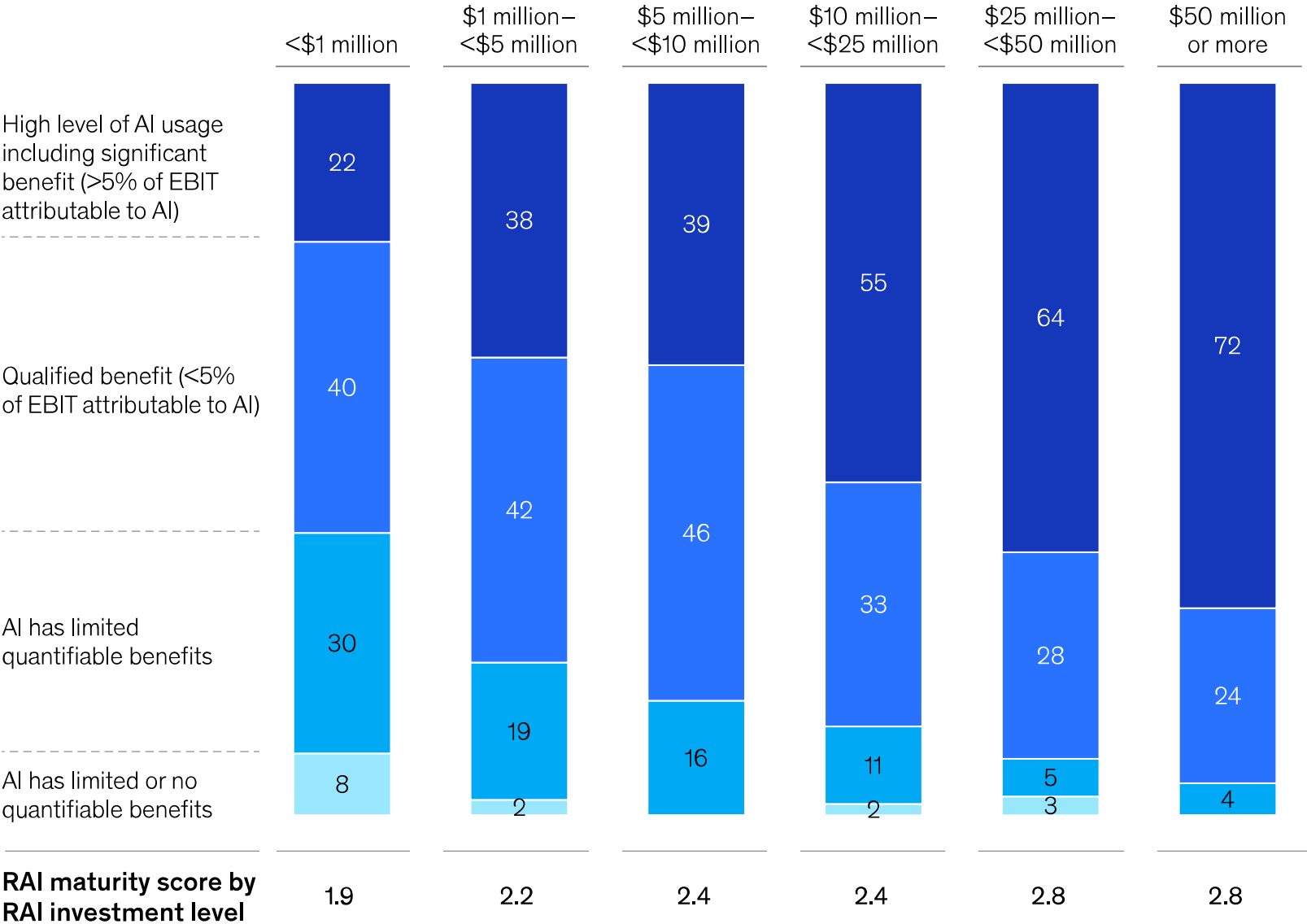
Insight 3: Investment in RAI is strongly associated with higher RAI maturity and realized value

Organizations investing \$25 million or more into RAI initiatives report significantly higher maturity scores and are far more likely to realize material AI benefits, including EBIT impact above 5 percent (Exhibit 5). This relationship reinforces that RAI investment is not a tax on innovation but a key enabler of sustained value creation.

Exhibit 5

Investment in responsible AI is associated with greater realized AI value and higher responsible AI maturity.

AI benefits to the organization by level of investment in responsible AI (RAI),¹
 % of respondents



Note: Figures may not sum to 100%, because of rounding.

¹Examples include hiring of RAI professionals; building or purchasing technical systems to comply with RAI principles; legal or professional services related to RAI; developing or integrating oversight, monitoring, and auditing tools; implementing workflow- or action-level safeguards and controls; conducting model evaluation, safety testing, or red-team exercises; upgrading security, access management, or data protection systems related to AI use; building and conducting training programs. "Unknown" RAI investment not shown (n = 21).

Source: McKinsey AI Trust Maturity Survey, Dec 2025–Jan 2026 (n = 496)

McKinsey & Company

Emerging risks: The key challenges organizations face today

The findings also explore the evolving risk landscape, including barriers to scaling agentic AI, the risks that organizations prioritize, and the disconnect between risk awareness, mitigation efforts, and incident preparedness.

Insight 4: Security and risk concerns are the top barrier to scaling agentic AI

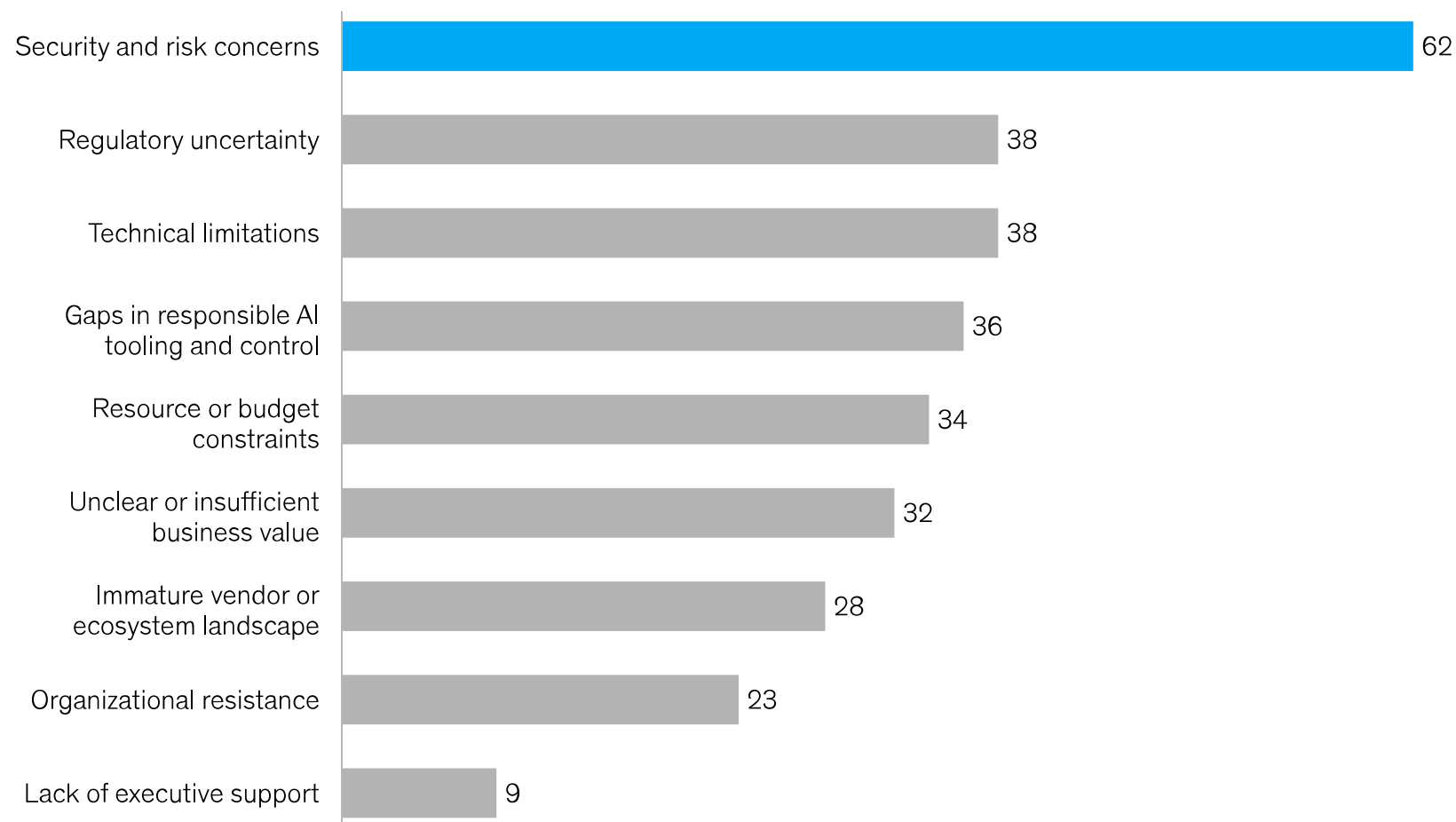
Nearly two-thirds of respondents cite security and risk concerns as the top barrier to fully scaling agentic AI, well ahead of regulatory uncertainty or technical limitations (Exhibit 6). This suggests that

organizations are less constrained by experimentation capabilities and more by confidence in their ability to safely deploy autonomous systems at scale.

Exhibit 6

Security and risk concerns are the most frequently cited obstacle to fully scaling agentic AI.

Main obstacles preventing organizations from reaching fully scaled agentic AI,¹ % of respondents



¹“None” (1%) and “Other” (2%) answer shares are not shown. Respondents were asked to choose all that apply.
Source: McKinsey AI Trust Maturity Survey, Dec 2025–Jan 2026 (n = 496)

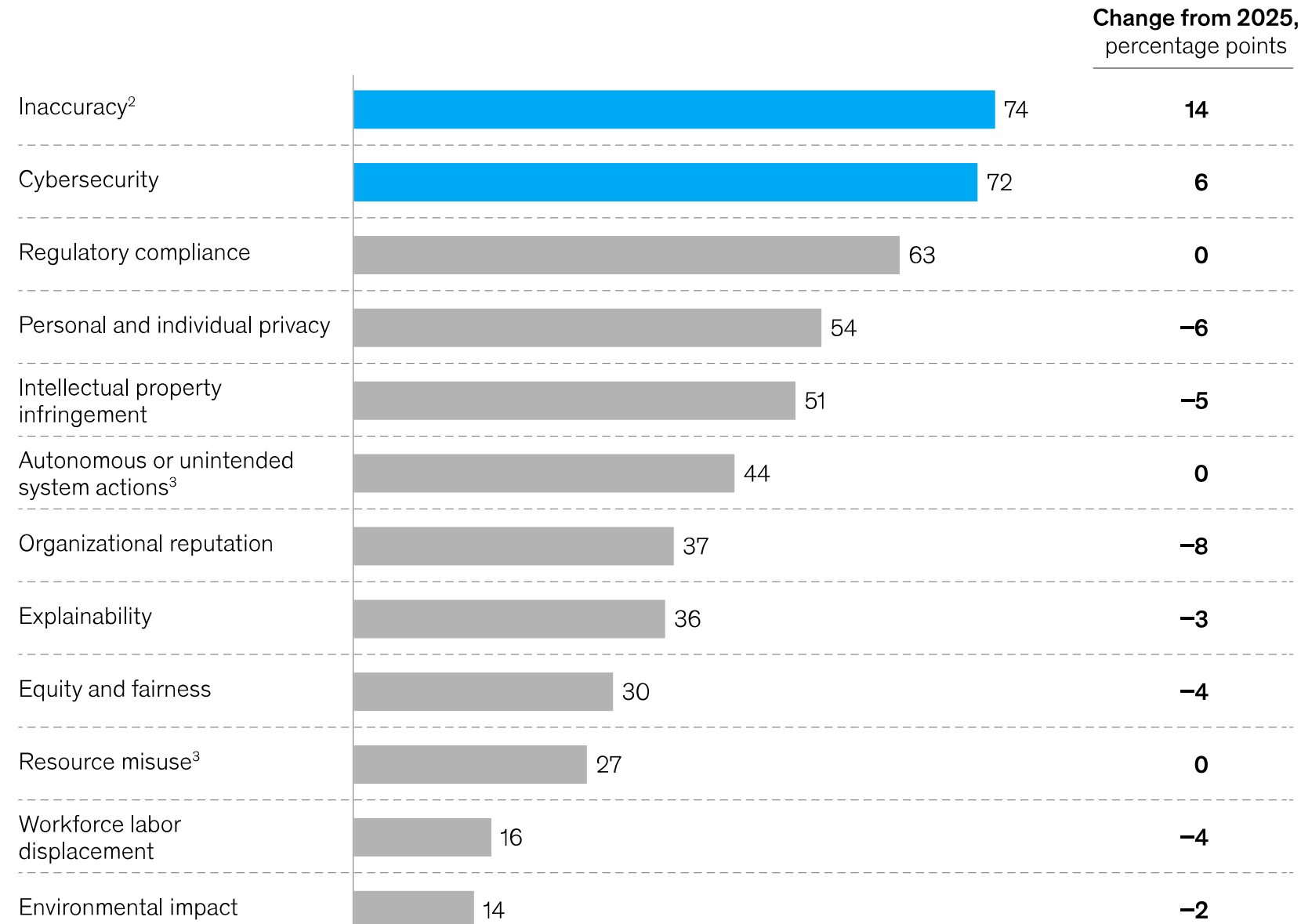
Insight 5: Inaccuracy and cybersecurity remain the most frequently cited AI risks as adoption expands

As AI adoption grows, 74 percent of respondents identify inaccuracy and 72 percent cite cybersecurity as highly relevant risks (Exhibit 7). These risks remain foundational concerns even as newer agentic risks emerge, highlighting that organizations must manage both traditional model risks and the expanded threat surface introduced by autonomy.

Exhibit 7

As AI adoption expands, inaccuracy and cybersecurity stand out as top-of-mind risks.

AI risks that organizations consider relevant and are working to mitigate,¹ % of respondents





¹“Other” (<1%) and “None” (<1%) answer shares are not shown. Respondents were asked to choose all that apply.

²Includes inconsistency and robustness failures.

³“Autonomous or unintended system actions” and “Resource misuse” were added in the 2026 AI Trust Maturity Survey (not included in 2025 survey).

Source: McKinsey AI Trust Maturity Survey, Dec 2024–Jan 2025 (n = 759); McKinsey AI Trust Maturity Survey, Dec 2025–Jan 2026 (n = 496)

McKinsey & Company

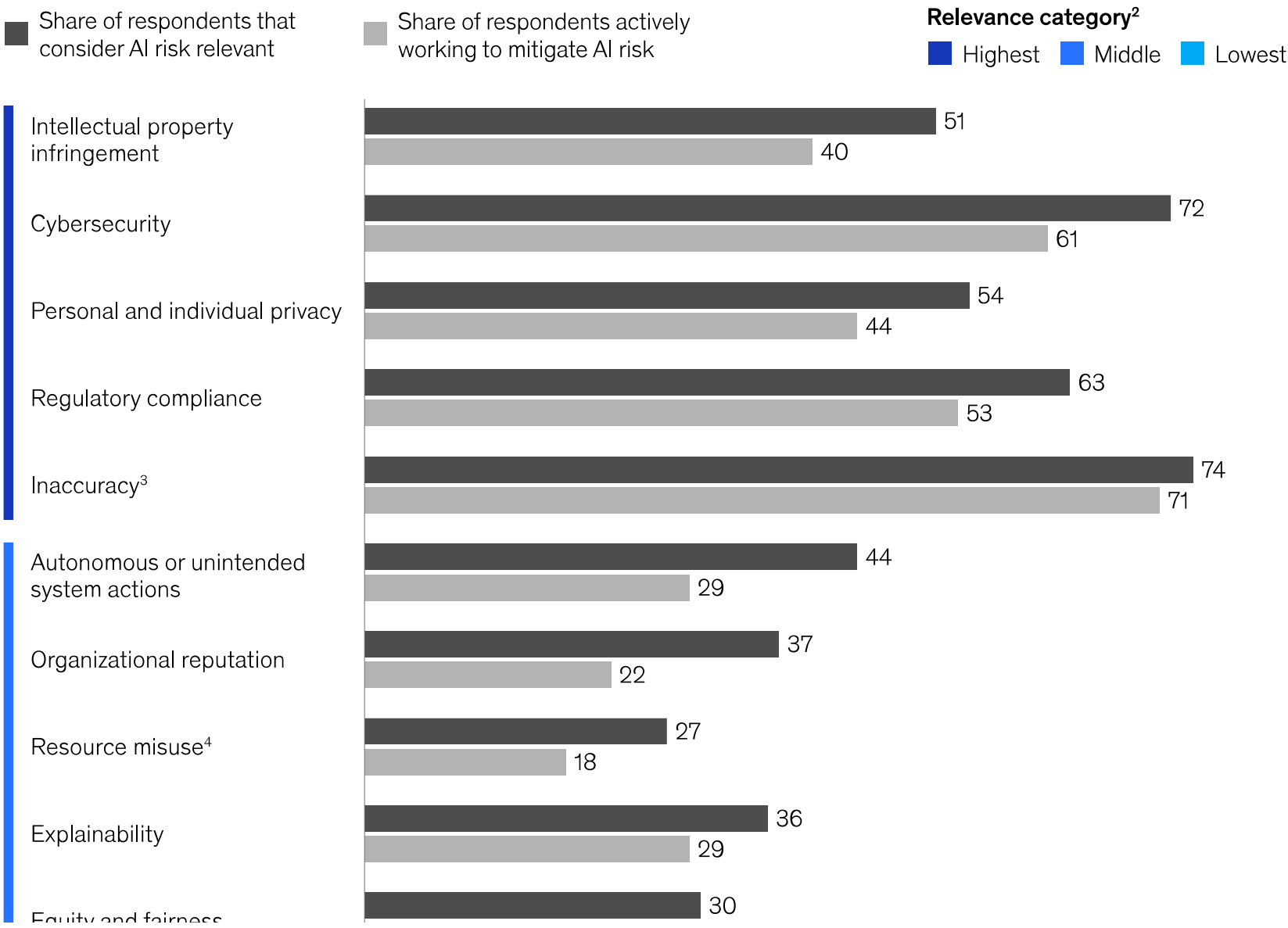
Insight 6: Active mitigation lags risk awareness across nearly every AI risk category

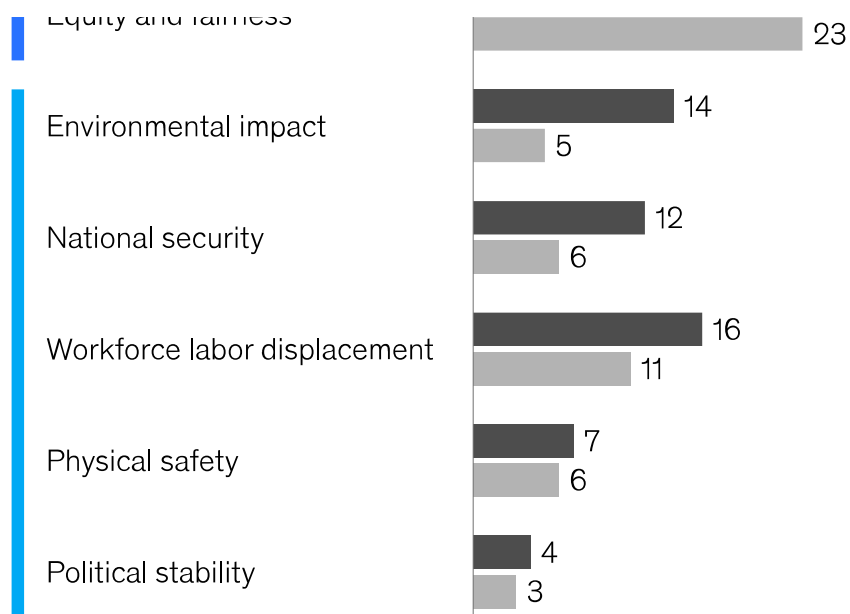
Across almost all risk types, respondents report a meaningful gap between the risks they consider relevant and those they are actively mitigating (Exhibit 8). This gap is especially pronounced for intellectual property infringement and personal privacy, suggesting that risk awareness is outpacing the implementation of controls, processes, and tooling needed to manage it effectively.

Exhibit 8

Active mitigation lags behind perceived relevance across nearly every risk category.

AI risk relevance and active work in mitigating risk,¹ % of respondents





¹“Other” (1% or less) and “None” (1% or less) answer shares are not shown. Respondents were asked to choose all that apply. ²Highest-ranked risk by relevance is defined as the top 5 risks by share of respondents that consider AI risk relevant; mid-ranked risk by relevance is defined as the middle 5 risks by share of respondents that consider AI risk relevant; lowest-ranked risk by relevance is defined as the bottom 5 risks by share of respondents that consider AI risk relevant. ³Includes inconsistency and robustness failures. ⁴“Autonomous or unintended system actions” and “Resource misuse” were added in the 2026 AI Trust Maturity Survey (not included in 2025 survey).

Source: McKinsey AI Trust Maturity Survey, Dec 2025–Jan 2026 (n = 496)

McKinsey & Company

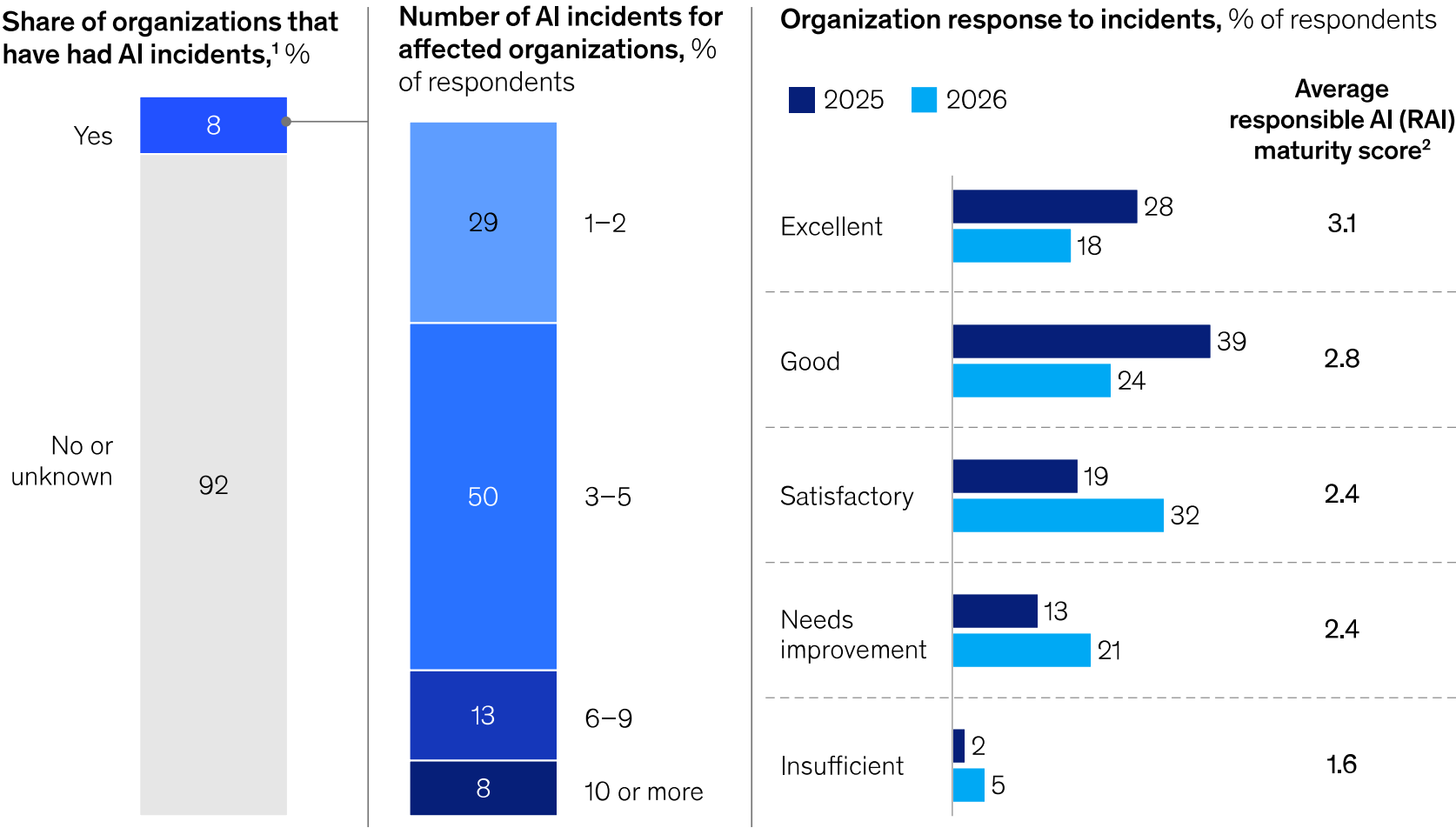
Insight 7: AI incident frequency remains stable, but confidence in organizational response has declined

The share of organizations reporting AI-related incidents has remained steady at roughly 8 percent, but perceptions of incident response quality have deteriorated. Almost 60 percent of respondents who

experienced incidents report satisfactory or negative views of their organization's response, indicating that while incidents may not be increasing, preparedness and response capabilities are failing to keep pace with growing system complexity (Exhibit 9).

Exhibit 9

The number of AI incidents remained constant since 2025, but confidence in the response to such episodes has declined.



¹According to the OECD, an AI incident is an event, circumstance, or series of events in which the development, use, or malfunction of one or more AI systems directly or indirectly leads to any of the following harms: injury or harm to the health of a person or groups of people; disruption of the management and operation of critical infrastructure; violations of human rights or a breach of obligations under the applicable law intended to protect fundamental, labor, and intellectual property rights; and harm to property, communities, or the environment.

²Average maturity calculated by taking the average of levels across organizations.

Source: McKinsey AI Trust Maturity Survey, Dec 2024–Jan 2025 (n = 759); McKinsey AI Trust Maturity Survey, Dec 2025–Jan 2026 (n = 496)

How enterprises and organizations are responding to emerging risks and challenges

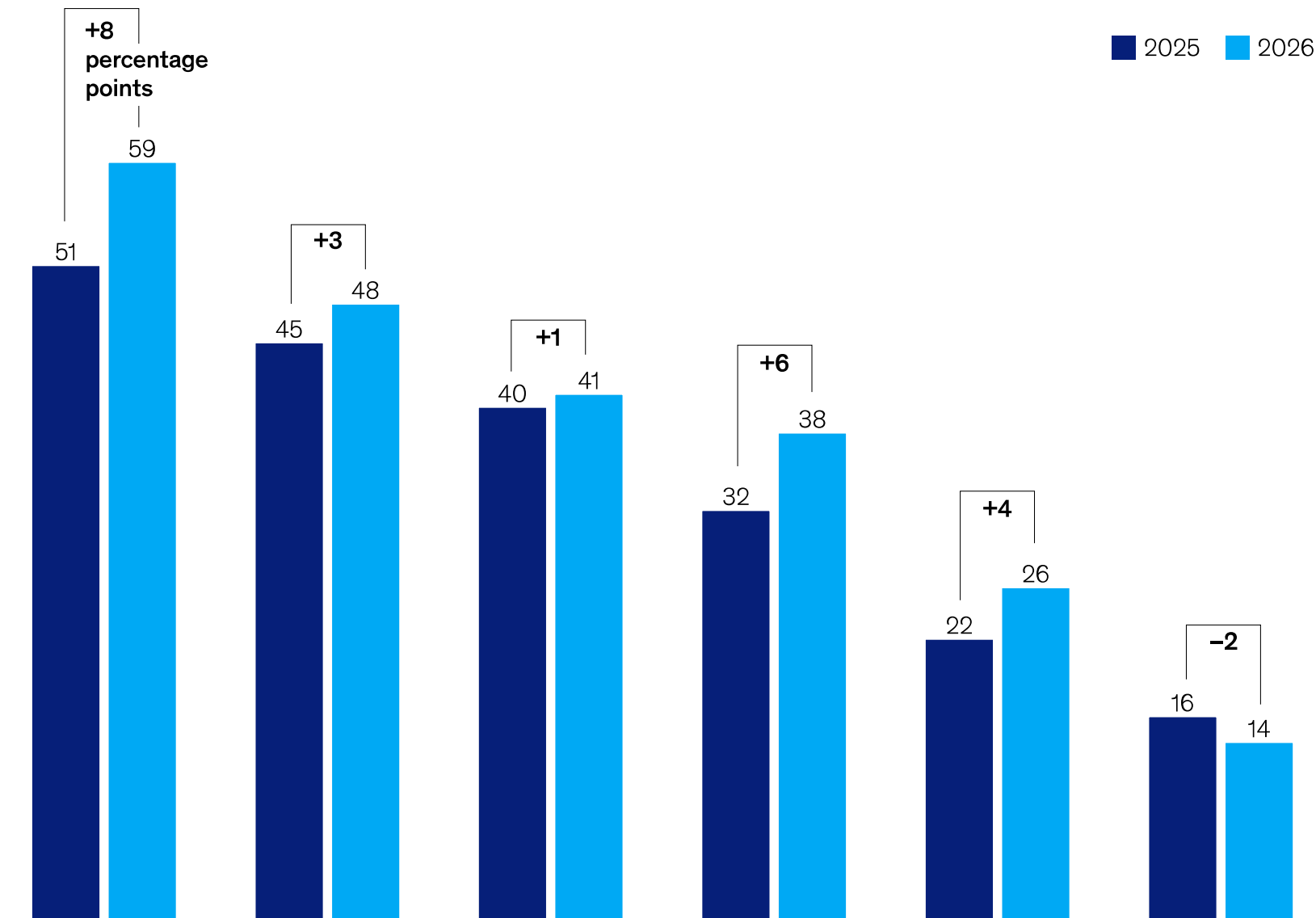
In response to emerging risks and challenges, organizations are working to strengthen the foundations of AI trust—closing capability gaps, clarifying accountability, and building the RAI capabilities needed to ensure trust accelerates innovation rather than constrain it.

Insight 8: Knowledge and training gaps are the leading barrier to RAI implementation

Nearly 60 percent of respondents cite knowledge and training gaps as the primary barrier to implementing RAI practices, up from about 50 percent last year (Exhibit 10). While executive support has improved, the data suggests that organizations continue to struggle with building the skills, awareness, and operational muscle required to embed RAI consistently across teams.

Knowledge and training gaps and resource and budget constraints are the largest barriers to implementing responsible AI measures.

Main obstacles preventing organizations from implementing responsible AI measures,¹
% of respondents



Knowledge and
training gaps

Resource or
budget constraints

Regulatory
uncertainty

Technical
limitations

Organizational
resistance

Lack of
executive support

¹“None” (2%) and “Unknown” (2%) answer shares not shown. Respondents were asked to choose all that apply.

Source: McKinsey AI Trust Maturity Survey, Dec 2024–Jan 2025 (n = 759); McKinsey AI Trust Maturity Survey, Dec 2025–Jan 2026 (n = 496)

McKinsey & Company

Insight 9: Organizations with explicit ownership for RAI have higher maturity than those without clear accountability

Organizations that assign clear ownership for RAI—particularly through AI-specific governance roles or internal audit and ethics teams—exhibit the highest average maturity levels, with an average score of 2.6. In contrast, organizations without a clearly accountable function lag behind materially (scoring an average of just 1.8), reinforcing the importance of explicit ownership and decision rights (Exhibit 11).

Insight 10: AI trust is increasingly viewed as a business enabler rather than a compliance exercise

Respondents report improvements in business outcomes, operational efficiency, and customer trust more frequently than negative outcomes (Exhibit 12). At the same time, the perceived influence of some regulatory frameworks has declined, suggesting a shift from compliance-led motivation toward value- and performance-driven adoption of AI trust.

Looking ahead: Trust as the enabler of scale

As AI systems become more autonomous and embedded in critical workflows, gaps in governance and risk management will become increasingly costly. Organizations that fail to establish clear accountability, robust controls, and effective monitoring mechanisms risk slower adoption, higher incident impact, and diminished stakeholder trust.

Conversely, organizations that treat AI trust as a core business capability, rather than as a compliance requirement, are better positioned to scale AI adoption to its full potential. These capabilities are not something that you can buy and install but rather require a concerted combination of policies, processes, people, and technology to build the foundations needed for agents, robots, and people to work together in new ways. Building a trustworthy innovation engine early will define those who capture long-term value from AI in the agentic era.

Gabriel Morgan Asaftei is a partner in McKinsey's New York office, where **Abby Sticha** is a consultant; **Roger Roberts** is a partner in the Bay Area office; and **Cécile Prinsen** is an associate partner in the London office.

How quantum technologies could rebalance the sustainability equation

Quantum technologies could help cut energy use of information and communications technology and unlock new climate and environmental solutions—if they are harnessed wisely.



By [Henning Soller](#)

Leads work on technology architecture and large-scale IT, data, and AI transformations across public- and private-sector institutions. Advises governments and enterprises on building scalable, sovereign-grade digital and AI platforms to enable resilient, future-ready ecosystems



By [Jess Fleming](#)

Serves R&D organizations in life sciences and chemicals on applying scientific AI and advanced analytics to accelerate innovation



By [Martina Gschwendtner](#)

Serves tech companies and their investors on incorporating gen AI, robotics, and quantum technologies

March 20, 2026 - The accelerating drumbeat of headlines heralding advances in AI is increasingly accompanied by a look at the broader impact of technology, [particularly on the environment](#). For example, the data centers at the heart of the AI revolution require a steady supply of power along with

millions of gallons of water a day to cool servers. Multiply that by the dramatic expansion of [data centers planned over the next several years](#), and the massive challenge that lies ahead starts to come into view.

Ironically, the same digital technologies that could increase resource demand also have the potential to reduce energy consumption. Used alongside existing classical infrastructure, quantum technologies could make certain simulations and optimizations far more efficient, although their overall impact on power use will depend on how and where they are deployed.

With the timeline for quantum hardware continuing to accelerate—recently published road maps suggest early, application-relevant systems emerging as soon as 2028¹—quantum players, enterprise end users, and investors should ensure they consider its impact on energy consumption. Here’s what they need to know.

Tech is a major contributor to climbing electricity usage

A look at trends over the past 15 years highlights the sharp, accelerating uptick in usage that is fueling electricity consumption. Since 2010, the number of internet users around the world has more than doubled, and global internet traffic, especially from mobile broadband, continues to rise rapidly; mobile broadband traffic alone crossed the one-zettabyte threshold in 2023.² Thanks to this increase, energy consumption by digital technologies rose by about 30 percent from 2007 to 2020 (Exhibit 1). This jump occurred despite major efficiency improvements in chips and data centers—a rebound effect known as

Jevons paradox, in which cheaper and more capable computing spurs higher overall demand and energy use.

Exhibit 1

Technology is a major contributor to climbing electricity usage.

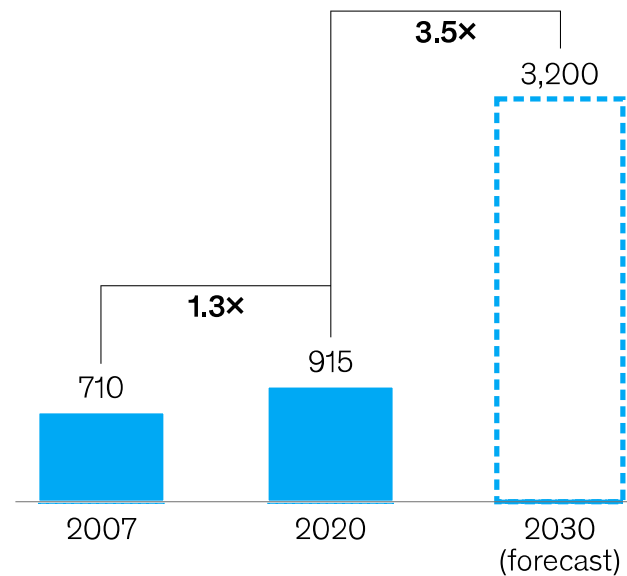
>2×

increase in the share of the global population using the internet, 2010–24 (from 29% to 71%)

>1 zettabyte

total global mobile broadband traffic in 2023

Information and communications technology (ICT) sector energy consumption, illustrative, terawatt-hours



Source: Ericsson; ITU World Telecommunication/ICT Indicators Database; Schneider Electric; World Bank

McKinsey & Company

The coming years could see the electricity consumption of information and communications technology triple, fueled by data-heavy traffic, AI, cryptocurrencies, data warehouses, and the continued adoption of

cloud services. The electricity demand from data centers alone could double by 2030, and power generation will have trouble keeping up.³ Indeed, 40 percent of existing AI data centers could be energy-constrained by 2027.⁴

The rise of quantum technologies

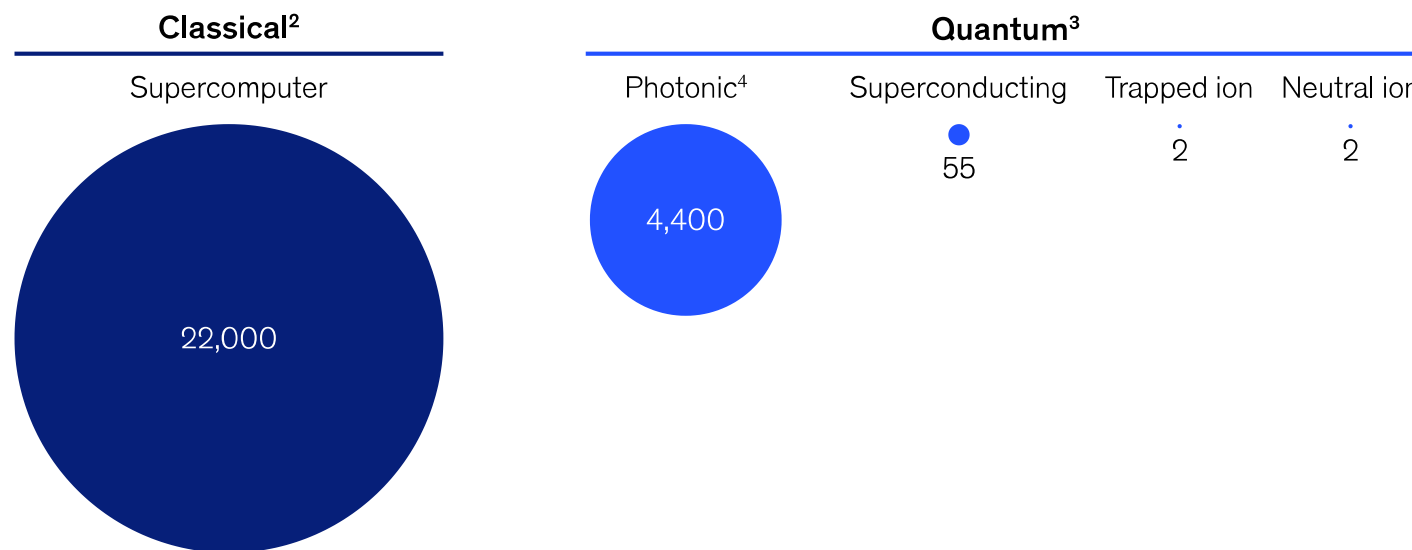
Interest and investment in quantum technologies continue to rise—for instance, governments have committed a cumulative \$42 billion to quantum projects as of 2024.⁵ Advancements in quantum infrastructure and hardware continue to accelerate, which will fuel rising energy consumption. The high energy needs for quantum hardware and software are continuous in daily operations and stem from several sources:

- GPUs that handle parallel processing are crucial for training AI and machine learning applications.
- Extensive cooling in data centers maintains constant and optimal hardware temperatures.
- High-performance servers run complex applications and process large volumes of data.
- Robust networking equipment manages vast amounts of transferred data.

Yet even with these electricity requirements, at the level of a single high-performance system, a large-scale quantum computer could draw far less power than an equivalent classical supercomputer for the same class of problem (Exhibit 2).

Despite high power requirements, a large-scale quantum computer could draw far less electricity than an equivalent classical supercomputer.

Indicative power requirements of classical and quantum computing architectures at scale,¹ kW



¹Figures are total electricity usage, including compute and other aspects such as cooling. ²Based on typical supercomputer GPU configurations and facility efficiency values. ³Based on bottom-up component model aggregating power draws for cryogenics, lasers, and control electronics, informed by current laboratory and vendor specifications. ⁴Sees the largest uncertainty (could vary by order of magnitude) and assumes large detector arrays cooled by cryocoolers, adjusted for plausible multiplexing advances.

In many architectures, particularly superconducting ones, the bulk of electricity consumption comes not from the quantum processors but from the infrastructure supporting them. These systems operate at millikelvin temperatures, maintained by dilution refrigerators that consume five to ten kilowatts (kW) each, with control and readout electronics adding several more kilowatts to total demand.⁵ However, alternative approaches show promise to substantially lower electricity consumption. Neutral-atom platforms rely on optical traps and room-temperature environments. A 1,000-qubit system of this kind can reportedly be operated with a total power draw of about ten kW.⁶

Given the rapid pace of hardware innovation and supporting systems, the energy use of quantum technologies will depend on not only the quantum processors but also the cooling, control electronics, and error correction required. Advances in hardware efficiency, [error correction](#), and algorithms now under active development could enable quantum computing to deliver significant performance gains with a much smaller environmental footprint.⁷ As quantum systems scale, locating them near abundant, reliable low-carbon electricity with sufficient grid capacity to support round-the-clock operation will become as critical as the hardware itself, mirroring the emerging infrastructure patterns seen with AI data centers.

How quantum technologies could shape electricity demand

Quantum technologies could be a key enabler in tackling climate change. While they will not be a magic bullet, they could make AI far more energy efficient per unit of useful computation—even as overall

demand for AI continues to grow—and support the development of solutions in other areas.

Quantum computing

Once deployed at scale, quantum computers could speed up certain computations while using less energy than classical hardware for those workloads, potentially reducing the energy required per AI task even if the total volume of compute continues to rise. For perspective, Meta aimed to operate about 1.3 million GPUs by the end of 2025, highlighting just how concentrated and resource-intensive today's AI workloads have become.⁸ Theoretically, a fully fault-tolerant quantum chip could one day rival the output of billions of classical GPUs on certain highly specialized problems such as simulating molecular quantum systems, which classical machines struggle to scale efficiently. In practical terms, that would mean plugging quantum processors into existing facilities as specialized accelerators that take over the most computationally intensive steps in selected AI and simulation workflows.

Quantum computing could also offer use cases for sustainability that were previously out of reach.

- *Quantum molecular simulation* could significantly improve the accuracy of modeling complex chemical reactions, such as those involved in carbon capture, battery chemistry, and hydrogen catalysis, enabling faster discovery and optimization of such techniques and their side effects. For example, more accurate modeling of the nitrogenase enzyme's structure and activity could be deployed to develop a green, scalable alternative for ammonia production.

- *Combinatorial optimization* involves finding solutions to more-complex problems (for example, load balancing in electrical grids for efficient integration of renewables), complementing the physical build-out of batteries and shiftable loads and flexible generation to make better use of available resources.
- *Quantum machine learning* could enable faster and more scalable analysis of large environmental data sets to improve climate and weather predictions, including hazardous events. This capability would enable earlier and more accurate interventions, reducing associated damage.

Many current superconducting systems require extreme cooling to be operational, which is highly energy inefficient.⁹ For quantum computers to operate with a minimal carbon footprint, they will require energy-efficient algorithms, optimized data centers (such as advanced cooling and dynamic power management), and low-carbon electricity supply.

Recent progress in quantum computing has been encouraging. Still, advancements in the next several years would be needed to achieve the performance and ease of implementation to support the widespread application of quantum computing and capture these ecological benefits.

Quantum sensing

Quantum sensors have the potential to offer more accurate, scalable environmental monitoring, far surpassing the limits of current technologies. These sensors could provide a number of advantages, such

as greater accuracy, higher sensitivity, and the ability to measure new territory (for example, within biological cells or in geographic regions without GPS).

The good news is that organizations are already exploring the deployment of quantum sensors in a range of climate tech use cases. For instance, they can rapidly detect hydrogen gas leaks, preventing potential hazards. BP led the HYDRI consortium, which developed quantum-based hydrogen sensors to enhance the safety of hydrogen gas.¹⁰ Similarly, the University of Birmingham and the British Geological Survey have explored the use of quantum sensors to monitor groundwater levels, aquifer health, and peatland regeneration.¹¹ The European Space Agency has used cold-atom quantum gravimeters to infer ice sheet dynamics over Iceland from ultraprecise gravity measurements.¹²

In addition, the deployment of quantum gravity sensors to monitor CO₂ movement in carbon capture and storage sites is being explored. A UK-led study used cold-atom gradiometers to model near-surface plumes, showing promise for detecting leaks with high sensitivity and low drift.¹³

These examples are emblematic of the promising developments in quantum sensors that could pave the way for broader, enduring impact.

Quantum communications

Cryptocurrency is energy-intensive, but its growing popularity has caused its adoption and activity to soar in recent years. Consider that Bitcoin's total trading volume reached \$9.1 trillion in March 2024, and its total annualized energy consumption is forecast to be 183 terawatt-hours.¹⁴

Quantum cryptography could drastically reduce the electricity demand associated with cryptocurrency mining, which requires extensive parallel hardware and high energy input. By using superposition to probe multiple states simultaneously, quantum systems offer computational shortcuts that consume far less electricity.

This year, researchers demonstrated a prototype blockchain that uses a new approach, “proof of quantum work.”¹⁵ Instead of relying on large numbers of conventional mining machines, which consume substantial amounts of electricity, this method employs quantum processors to complete the computational steps needed to validate transactions. If scaled, it offers a pathway to reduce the environmental impact of digital asset infrastructure while maintaining security standards. In addition, Toshiba Europe successfully transmitted quantum-secure communications over 250 kilometers of live telecom fiber using components that operate at room temperature.¹⁶ This approach eliminates the need for energy-intensive cooling, making it easier and more efficient to deploy quantum-secure networks on existing infrastructure.

As noted, the coming years could see progress on a number of related fronts. Investments in AI and data centers are forecast to significantly increase electricity use, which is already raising concerns about supply and connectivity capacity. At the same time, advances could make quantum technologies less energy-intensive, with even high-demand superconducting systems projected to require less power than classical supercomputers. And quantum applications could directly enhance sustainability across sectors.

The intersection of these parallel tracks has the potential to bend the energy consumption curve. Notably, quantum technologies won't solve climate challenges on their own, but they could help reduce the energy use of digital technologies while unlocking new climate solutions. Their impact will hinge on design (such as advances in cooling and error correction) and siting (the potential of renewable energy sources to power data centers).

So, what actions can stakeholders take now? Given the accelerated development timeline for quantum technologies, companies could integrate future quantum energy demands into procurement and infrastructure plans. The projected growth of quantum technologies—\$97 billion in global revenues by 2035—means investors would do well to incorporate this lens into portfolio company road maps. And because the cloud will be the primary access point for quantum technologies in the long term for most industry players, security and data sovereignty issues will become a higher priority.

Companies could get ahead of these issues by assessing the location of providers and their controls and governance related to data.¹⁷ Careful consideration of potential implications can ensure that all players are well prepared to fully harness quantum technologies.

Henning Soller is a partner in McKinsey's Frankfurt office, **Jess Fleming** is a data scientist in the London office, and **Martina Gschwendtner** is a consultant in the Munich office.

The authors wish to thank Duc Nam Nguyen, Victor Kermans, and Waldemar Svejstrup for their contributions to this blog post.

¹ Sophia Chen, "IBM aims to build the world's first large-scale, error-corrected quantum computer by 2028," *MIT Technology Review*, June 10, 2025; Miranda Davis, Daniela Sirtori, and Isabella Ward,

“Chicago’s \$1 billion quantum computer set to go live in 2028,” Bloomberg, July 21, 2025.

² “Two-thirds of the world’s population uses the Internet, but 2.7 billion people remain offline,” ITU, 2022.

³ “AI is set to drive surging electricity demand from data centres while offering the potential to transform how the energy sector works,” IEA, April 10, 2025.

⁴ “Gartner predicts power shortages will restrict 40% of AI data centers by 2027,” Gartner, November 12, 2025.

⁵ “Dilution refrigerator: Everything you need to know [2025],” SpinQ, May 30, 2025.

⁶ “Quantum computing to greener calculation,” Pasqal, October 5, 2023.

⁷ Marco Fellous-Asiani et al., “Optimizing resource efficiencies for scalable full-stack quantum computers,” arXiv, updated October 16, 2023.

⁸ Jaspreet Singh, “Meta to spend up to \$65 billion this year to power AI goals, Zuckerberg says,” Reuters, January 24, 2025.

⁹ *Q Blog*, “Could quantum computing help curb AI’s carbon footprint?,” blog entry by Gabriella Skoff, Project Q University of Sydney, June 21, 2019.

¹⁰ “HYDRI-HYDrogen sensor for Industry,” UK Research and Innovation, updated on January 8, 2026.

¹¹ “Researchers awarded funding to use quantum sensor technology for environmental applications,” University of Birmingham, March 20, 2023.

¹² “New campaign dataset for Airborne Quantum Gravimetry,” European Space Agency, July 16, 2024.

¹³ “Feasibility study into quantum technology-based gravity sensing for carbon dioxide capture and storage,” UKCCSRC, March 2022.

¹⁴ As of February 17, 2026. Nasdaq and CoinDesk data; Ibrahim Ajibade, “Crypto market Q1 2024 round-up: Trading volume hits \$9.1 trillion as despite market dip,” Nasdaq, April 5, 2024; “Bitcoin network power

demand,” Cambridge Electricity Consumption Index, accessed February 17, 2026.

¹⁵ Mohammad H. Amin et al., “Blockchain with proof of quantum work,” arXiv, updated on February 9, 2026.

¹⁶ “Toshiba breakthrough brings quantum communications to existing national-scale telecommunications infrastructure,” Toshiba, April 24, 2025.

¹⁷ Justin Coupel and Tasnuva Farheen, “Security vulnerability in quantum cloud systems: A survey on emerging threats,” arXiv, April 27, 2025.